DATA STRUCTURES

MODEL QUESTION PAPER • MCA I SEMESTER

Exam: Continuous Assessment Test **Maximum Marks:** 50 **Time:** 90 Minutes

Note: This is a practice prediction prepared from the provided Data Structures question bank and the previous question-paper pattern. It is not the confirmed exam paper.

PART A — TWO-MARK QUESTIONS

Answer all questions. (10 × 2 = 20 Marks)

1. Define a Data Structure. Give two examples.

2. What is an Abstract Data Type (ADT)? Give one example.

3. Define time complexity and space complexity.

4. Differentiate between Big-O and Big-Ω notations.

5. What is a linked list? State one advantage over an array.

6. Differentiate between a singly linked list and a doubly linked list.

7. Define a stack and explain the LIFO principle.

8. What are PUSH and POP operations?

9. Define a queue and explain the FIFO principle.

10. What is a circular queue? State one advantage over a linear queue.

PART B — FIFTEEN-MARK QUESTIONS

Answer either (a) or (b) in each question. (2 × 15 = 30 Marks)

11. UNIT I — DATA STRUCTURES AND LINKED LISTS

11(a) Explain the fundamental concepts, types, classification and applications of data structures with suitable examples.

OR

11(b) Explain singly linked lists with a neat diagram. Describe the algorithms for insertion and deletion at the beginning, end and specified position.

12. UNIT II — STACK AND QUEUE

12(a) Explain the basic stack operations PUSH and POP with suitable algorithms and examples.

OR

12(b) Explain the implementation of a circular queue. Describe the ENQUEUE and DEQUEUE operations with suitable algorithms and examples.

REVISION PRIORITY

1. Singly Linked List — insertion and deletion (Unit I)

2. Stack — PUSH and POP operations (Unit II)

3. Circular Queue — ENQUEUE and DEQUEUE (Unit II)

4. Types and classification of Data Structures (Unit I)

5. Abstract Data Type (ADT) (Unit I)

6. Big-O, Big-Ω and Big-Θ notations (Unit I)

7. Arrays vs Linked Lists (Unit I)

8. Infix to Postfix conversion using Stack (Unit II)

9. Doubly Linked List and Circular Linked List (Unit I)

10. Linear Queue vs Circular Queue (Unit II)

Priority is a study suggestion based on the supplied question bank, not a guaranteed prediction of the exam.

ADDITIONAL TWO-MARK REVISION

1. What is an algorithm?

2. What are the characteristics of an algorithm?

3. What is asymptotic analysis?

4. What is the time complexity of accessing an array element?

5. What is the purpose of a head pointer?

6. What is the worst-case time complexity of searching a singly linked list?

7. What are stack overflow and stack underflow?

8. Convert A + B * C into postfix notation.

9. What are the two ends of a queue?

10. Define priority queue.

11. Define deque.

12. Mention any two applications of stacks.

13. Mention any two applications of queues.

15-MARK ANSWER-WRITING FORMAT

Definition • Introduction • Neat diagram • Types/classification • Components • Operations • Algorithm • Example • Advantages • Disadvantages • Applications • Conclusion

Tip: Include a diagram, algorithm and example for linked-list, stack and queue questions wherever relevant. Write only the sections that fit the question.